

SOLAR FILAMENT ERUPTION & SATELLITE RADIATION ALERT

OFFICIAL SPACE ENVIRONMENT TELEMETRY BULLETIN | ISSUED: 2026-08-21 11:18 UTC

1. EXECUTIVE EVENT SUMMARY

SOURCE REGION : N37.6° W0.0° (Stonyhurst)
ERUPTION RISK : 24h: 18.4% | 48h: 33.4% [MODERATE WATCH]
PROJECTED FLARE : C-Class Sub-Flare Potential Soft X-Ray Class
EARTH CONNECTIVITY: Magnetic Separation $\Delta\phi = 1.2^\circ$ (High-Flux Parker Spiral Path)
ASTRONAUT SAFETY : STANDARD SPACE ENVIRONMENT PROCEDURES MAINTAINED

2. DETECTED FILAMENT MORPHOMETRIC MEASUREMENTS

- True Physical Spine Length : 72,500 km (0.104 R_Sun)
- True Plasma Surface Area : 1.08e+08 km²
- Magnetic Neutral Line Shear: 45.0° (High Free Energy Polarity Inversion Line)
- Mean Chromospheric Contrast: 0.000 $\Delta I/I_0$

3. CORONAL MASS EJECTION (CME) DRAG-BASED IMPACT FORECAST

CME LAUNCH SPEED : 550.0 km/s (Estimated Initial Plasma Ejection)
1-AU ARRIVAL SPEED: 479.6 km/s (Aerodynamically Decelerated)
TRANSIT TIME : 81.9 Hours (3.4 Days)
ESTIMATED IMPACT : 2026-08-24 21:12 UTC
GEOMAGNETIC SCALE : NOAA G1 (Minor) | Kp Index: 5.0/9
POWER GRID IMPACT : Intermittent voltage regulation alarms; induced pipeline currents

4. OPERATIONAL AEROSPACE MISSION DIRECTIVES

- Human Spaceflight (ISS / Tiangong) : Restrict crew to radiation-hardened service modules.
- Lagrange Sentinels (DSCOVR / JWST) : Throttle optical sensors; enable radiation scrubbing on SSR.
- GNSS Navigation (GPS III / Galileo): Upload clock phase corrections; monitor L-band scintillation.
- LEO Mega-Constellations (Starlink) : Orient satellites edge-on (knife-edge) to mitigate drag decay.

Generated by Solar Filament AI & Space Weather Intelligence Platform (2-Stage Coarse-to-Fine Pipeline)